

Lecture #12: Solving the Linear Programming Problem, continued

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Math 190: Mathematics of Sustainability, Emory University

March 2, 2026



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Refresher from last class

Definition (End Points, Interior Points)

The line segment joining $\mathbf{x}_1$ and $\mathbf{x}_2$ is defined as

$$\{\mathbf{x} \in \mathbb{R}^n | \mathbf{x} = p\mathbf{x}_1 + (1 - p)\mathbf{x}_2\}, \quad 0 \leq p \leq 1$$

We define the *end points* as

Remark



Making it more general in higher dimensions

Definition

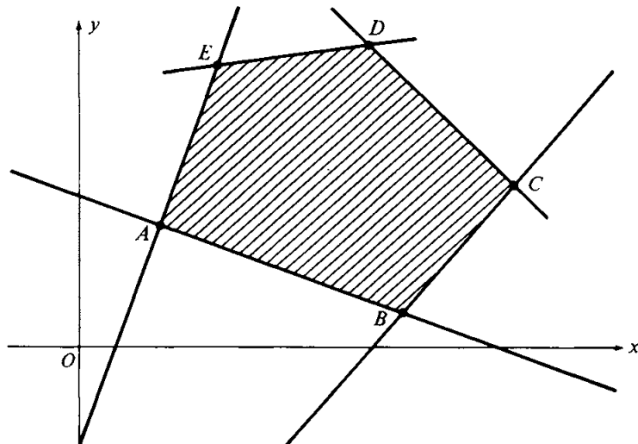
A point $\mathbf{x} \in \mathbb{R}^n$ is a *convex combination* of the points $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r$ in $\mathbb{R}^n$ if for some real numbers $c_1, c_2, \dots, c_r$ satisfy

Definition

A point $x \in C$ is an *extreme point* if it



Visualizing Extreme Points



Another Refresher

Definition

A *bounded* convex set is



The Fundamental Theorem of Linear Programming

Theorem (Fundamental Theorem of LP)

Let S be the set of feasible solutions to a general linear programming problem.

- ① If S is nonempty and bounded,*
- ② If S is nonempty and not bounded*
- ③ If an optimal solution to the problem does not exist,*



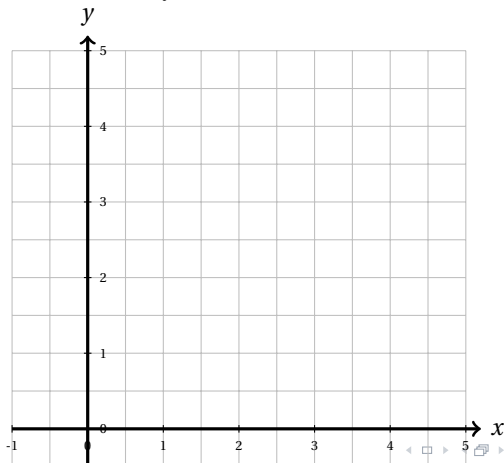
Consider the linear programming problem

maximize $z = 120x + 100y$ subject to

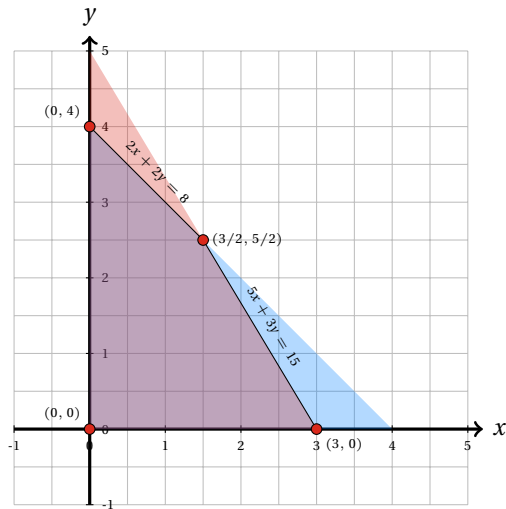
$$2x + 2y \leq 8$$

$$5x + 3y \leq 15$$

$$x, y \geq 0$$



The convex set of all feasible solutions are shown below:



We see that the extreme points of the convex set S are in red above: $(0, 0)$, $(3, 0)$, $(0, 4)$, and $(\frac{3}{2}, \frac{5}{2})$.

Extreme point (x, y)	Value of objective function $z = 120x + 100y$
$(0, 0)$	
$(3, 0)$	
$(0, 4)$	
$(\frac{3}{2}, \frac{5}{2})$	



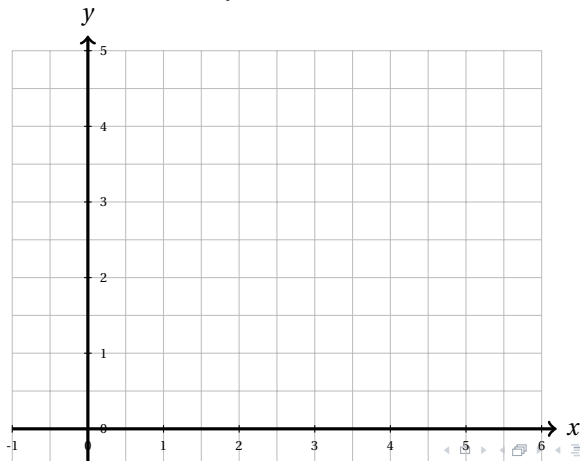
Consider the linear programming problem

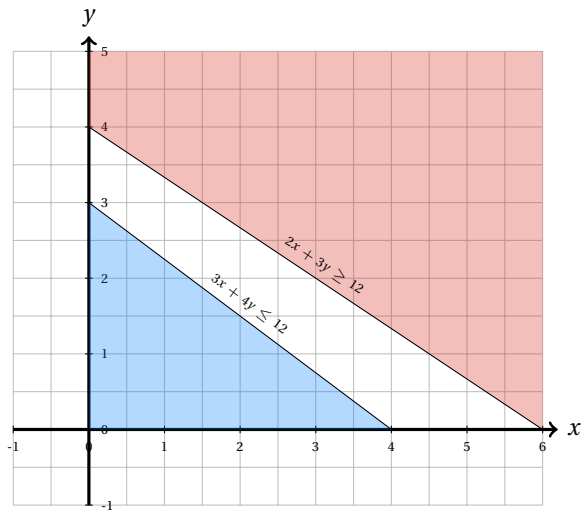
maximize $z = 2x + 5y$ subject to

$$2x + 3y \geq 12$$

$$3x + 4y \leq 12$$

$$x, y \geq 0$$





What does the FTLP tell us?

Theorem (Fundamental Theorem of LP)

Let S be the set of feasible solutions to a general linear programming problem.

- ① If S is nonempty and bounded, then an optimal solution to the problem exists and occurs at an extreme point.*
- ② If S is nonempty and not bounded and if an optimal solution to the problem exists, then an optimal solution occurs at an extreme point.*
- ③ If an optimal solution to the problem does not exist, then either S is empty or S is unbounded.*



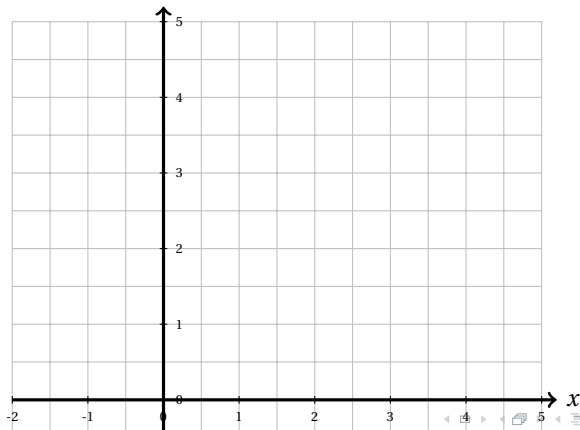
Another Example

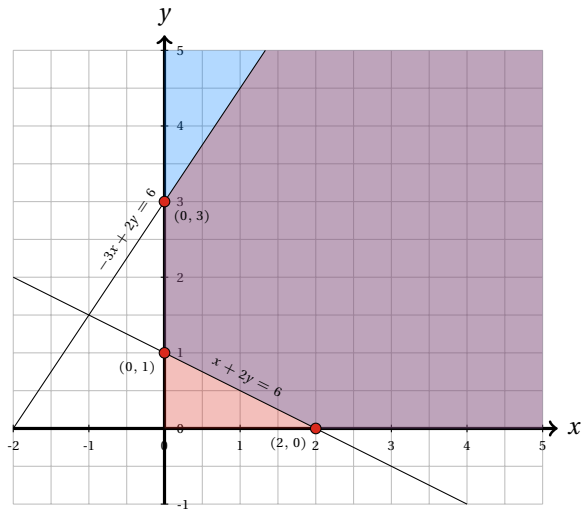
minimize $z = 2x + 5y$ subject to

$$-3x + 2y \leq 6$$

$$x + 2y \geq 2$$

$$x, y \geq 0$$





There are many points that are in the feasible solution. But what does the FTLP tell us?

Theorem (Fundamental Theorem of LP)

Let S be the set of feasible solutions to a general linear programming problem.

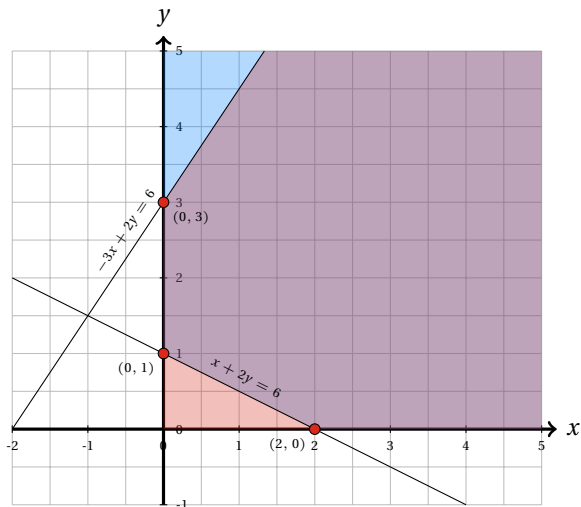
- ① *If S is nonempty and bounded, then an optimal solution to the problem exists and occurs at an extreme point.*
- ② *If S is nonempty and not bounded and if an optimal solution to the problem exists, then an optimal solution occurs at an extreme point.*
- ③ *If an optimal solution to the problem does not exist, then either S is empty or S is unbounded.*



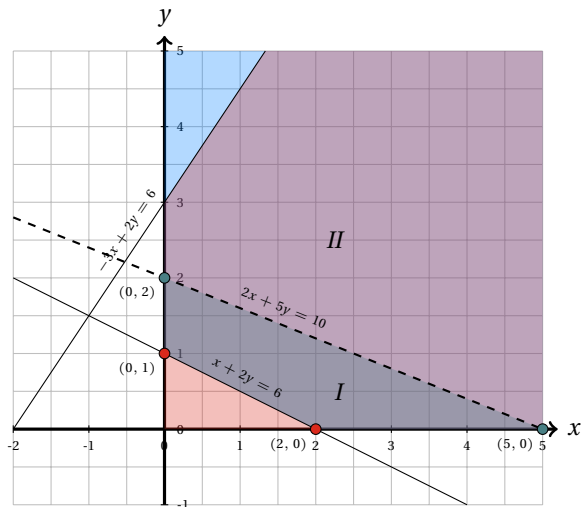
Extreme point (x, y)	Value of objective function $z = 2x + 5y$
$(0, 3)$	
$(0, 1)$	
$(2, 0)$	



Another Way of Solving this Problem



Making a bounded domain Problem



This means that we now have a bounded, nonempty solution region, which means that there must be an optimal solution. We can then fill out the table below:

Extreme point (x, y)	Value of objective function $z = 2x + 5y$
$(0, 2)$	
$(5, 0)$	
$(0, 1)$	
$(2, 0)$	



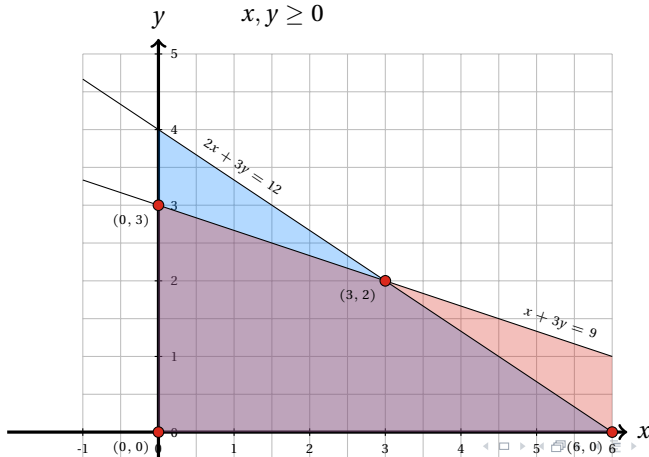
Last Example

maximize $z = 2x + 3y$ subject to

$$x + 3y \leq 9$$

$$2x + 3y \leq 12$$

$$x, y \geq 0$$



This means that we now have a bounded, nonempty solution region, which means that there must be an optimal solution. We can then fill out the table below:

Extreme point (x, y)	Value of objective function $z = 2x + 3y$
$(0, 0)$	
$(0, 3)$	
$(6, 0)$	
$(3, 2)$	



Using this method, we see that both extreme points of $(6, 0)$ and $(3, 2)$ give the maximal value of 12. We also know that the line segment joining these two points are

For any point (x, y) on this line segment, we have the value

